Question 1 : Define Power BI and What are the key components of the Power BI ecosystem? Briefly explain: ● Power BI Desktop ● Power BI Service ● Power BI Mobile ● Power BI Gateway

Answer: Power BI Overview
Power BI is a business intelligence (BI) and data visualization tool developed by Microsoft. It enables users to connect to various data sources, transform raw data into meaningful insights, and share interactive dashboards and reports. Its ecosystem is designed to support data analysis, reporting, and collaboration across devices and platforms.

Key Components of the Power BI Ecosystem
1. Power BI Desktop
A free Windows application used for creating reports and dashboards.

Provides powerful data modeling, transformation, and visualization capabilities.

Users can connect to multiple data sources, clean and shape data using Power Query, and build interactive reports.

It is mainly used by data analysts and developers for report creation before publishing to the Power BI Service.

2. Power BI Service (Power BI Online)
A cloud-based platform where reports and dashboards created in Power BI Desktop can be published and shared.

Enables collaboration, sharing, and real-time access to reports across teams and organizations.

Provides features like scheduled data refresh, role-based security, and integration with Microsoft 365.

Ideal for business users to consume and interact with reports without needing to design them.

3. Power BI Mobile
Mobile applications available for iOS, Android, and Windows devices.

Allows users to access and interact with dashboards and reports on the go.

Provides a responsive and touch-friendly interface for viewing insights anytime, anywhere.

Useful for executives and managers who need quick access to KPIs and business metrics.

4. Power BI Gateway
A bridge that connects on-premises data sources with Power BI Service.

Ensures secure data transfer between local databases (like SQL Server, Oracle, etc.) and the cloud.

Two types:

Personal Gateway: For individual use, refreshes data from local sources.

Enterprise Gateway: For organizational use, supports multiple users and scheduled refreshes.

Essential for organizations that rely on on-premises data but want to leverage cloud-based reporting.

Question 2 : Compare the following Power BI visuals:
● Pie Chart vs Donut Chart
● Bar Chart vs Column Chart When would you prefer one over the other? Give one example for each pair.

Answer: # Pie Chart vs Donut Chart

- Pie Chart
    - Displays data as slices of a circle.
    - Good for showing **proportions** of categories in a whole.
    - Simple and widely recognized.
- Donut Chart
    - Similar to a pie chart but with a hole in the center.
    - Allows space in the middle for labels or KPIs.
    - Visually cleaner when comparing multiple categories.

When to prefer:

- Use Pie Chart when you want a straightforward view of category proportions.
- Use Donut Chart when you want to highlight a central metric (like total sales) along with category breakdown.

Example:

- Pie Chart → Showing the percentage of students in different departments.
- Donut Chart → Showing sales by product category with total revenue displayed in the center.

Bar Chart vs Column Chart

- Bar Chart
    - Horizontal bars.

    - Best for comparing categories with long names or when you want easy readability.
- Column Chart
    - Vertical bars.
    - Best for showing time-based trends or sequential data.

When to prefer:

- Use Bar Chart when category names are lengthy or when comparison is the focus.
- Use Column Chart when showing changes over time or progression.

Example:

- Bar Chart → Comparing revenue across different countries (names are long).
- Column Chart → Showing monthly sales growth over a year.

Question 3 : Explain the significance of: ● Star schema vs Snowflake schema ● Primary key vs Foreign key in relationships (Power BI) Why is cardinality important?

Answer:

Star Schema vs Snowflake Schema

- Star Schema
    - Structure: Central fact table linked directly to dimension tables.
    - Simple, easy to understand, faster query performance.
    - Best for reporting and dashboards in Power BI.
- Snowflake Schema
    - Structure: Dimension tables are normalized into multiple related tables.
    - More complex, reduces data redundancy, but queries can be slower.
    - Useful when dimensions have hierarchical relationships (e.g., Country → State → City).

Significance:

- Star schema is preferred for **performance and simplicity**.
- Snowflake schema is used when **data integrity and normalization** are more important.

Primary Key vs Foreign Key (Power BI Relationships)

- Primary Key
    - A unique identifier in a table (e.g., CustomerID in Customers table).
    - Ensures each row is distinct.
- Foreign Key
    - A column in another table that references the primary key (e.g., CustomerID in Orders table).

    - Creates relationships between tables.

Significance:

- Primary and foreign keys allow Power BI to link tables and build a relational data model.
- Without these relationships, cross-table analysis (like customer sales) isn't possible.

Why Cardinality is Important

- Cardinality = Defines the relationship type between tables:
    - One-to-One (1:1) → Each record in Table A matches one record in Table B.
    - *One-to-Many (1:)** → One record in Table A relates to multiple records in Table B (most common).
    - *Many-to-Many (*:)** → Multiple records in both tables relate to each other.

Significance:

- Cardinality ensures Power BI interprets relationships correctly.
- Wrong cardinality can lead to duplicate counts, incorrect aggregations, or misleading visuals.
- Example: Customer (1) → Orders (Many). If set incorrectly, sales totals could be inflated.

Question 4 : Differentiate between: ● Calculated column vs Measure Also, define Row context and Filter context with simple examples.

Answer: Calculated Column vs Measure

- Calculated Column
    - Created at the row level in a table.
    - Uses DAX formulas to add new data fields.
    - Stored in the data model (takes up memory).
    - Evaluated during data refresh.
    - Example: Adding a column `Profit = Sales - Cost` for each row.
- Measure
    - Calculated on the fly when used in visuals.
    - Uses DAX to perform aggregations (SUM, AVERAGE, COUNT, etc.).
    - Not stored in the table, only computed when needed.
    - More efficient for large datasets.
    - Example: `Total Sales = SUM(Sales[Amount])`.

Key Difference:

- Calculated Column → row-by-row, stored in the model.
- Measure → dynamic, aggregated, calculated at query time.

Row Context vs Filter Context

- Row Context
  - Exists when a calculation is evaluated for each row in a table.
  - Common in calculated columns.
  - Example: In a calculated column `Profit = Sales - Cost`, Power BI evaluates the formula for each row individually.
- Filter Context
  - Exists when filters are applied to data (from slicers, visuals, or DAX functions).
  - Common in measures.
  - Example: A measure `Total Sales = SUM(Sales[Amount])` will return different results depending on filters like “Region = Asia” or “Year = 2025”.

Question 5: What is the difference between a report and a dashboard in Power BI?

Answer: # Power BI Report

- **Definition:** A multi-page, detailed view of data created in Power BI Desktop and published to the Service.
- Features:
  - Can contain multiple pages.
  - Rich visuals (charts, tables, maps, slicers).
  - Highly interactive with drill-through and filtering.
- Use Case: Best for in-depth analysis and exploration of datasets.

Example: A sales report showing revenue trends, customer demographics, and product performance across multiple pages.

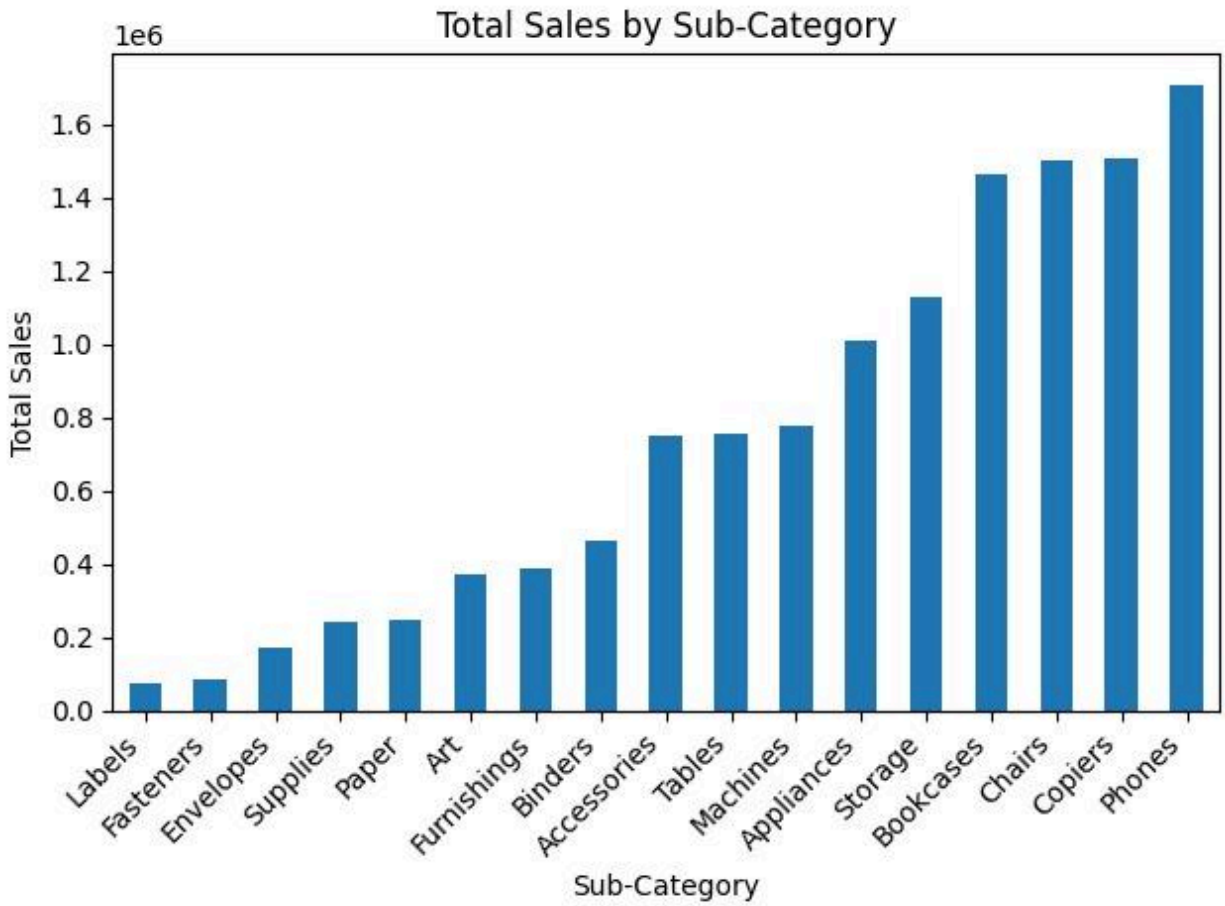
Power BI Dashboard

- Definition: A single-page, consolidated view created in the Power BI Service.
- Features:
  - Combines visuals (tiles) from multiple reports or datasets.
  - Focused on key metrics and KPIs.
  - Provides a quick, high-level snapshot.
- Use Case: Best for executive monitoring and quick decision-making.

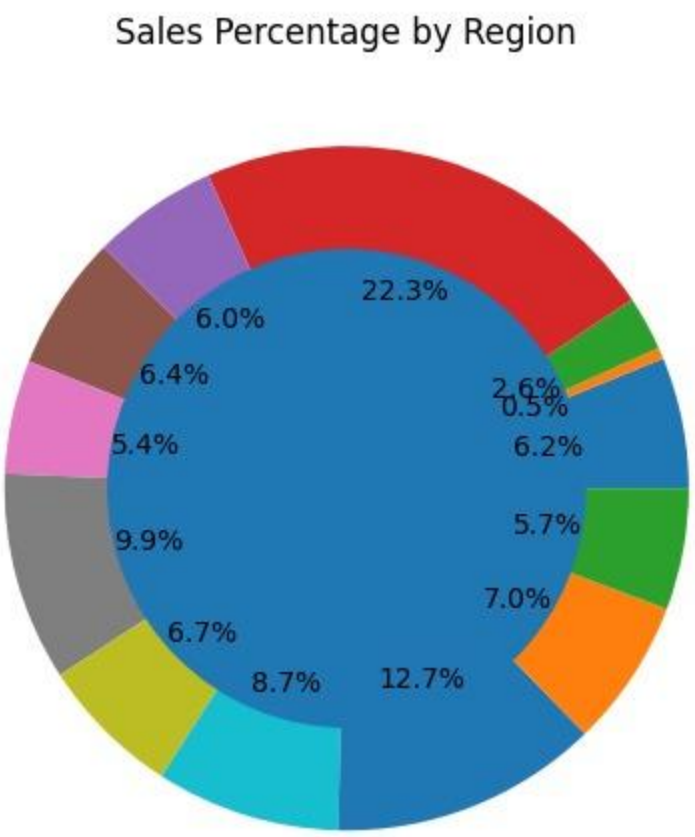
Example: A dashboard showing total revenue, top-selling products, and regional performance in one glance.

Question 6 :Using the Sample Superstore dataset: ● Create a Clustered Bar Chart to display Total Sales by Sub-Category ● Create a Donut Chart for Sales % by Region Provide screenshots of both visuals. DATASET LINK : Global_superstore2

Answer:



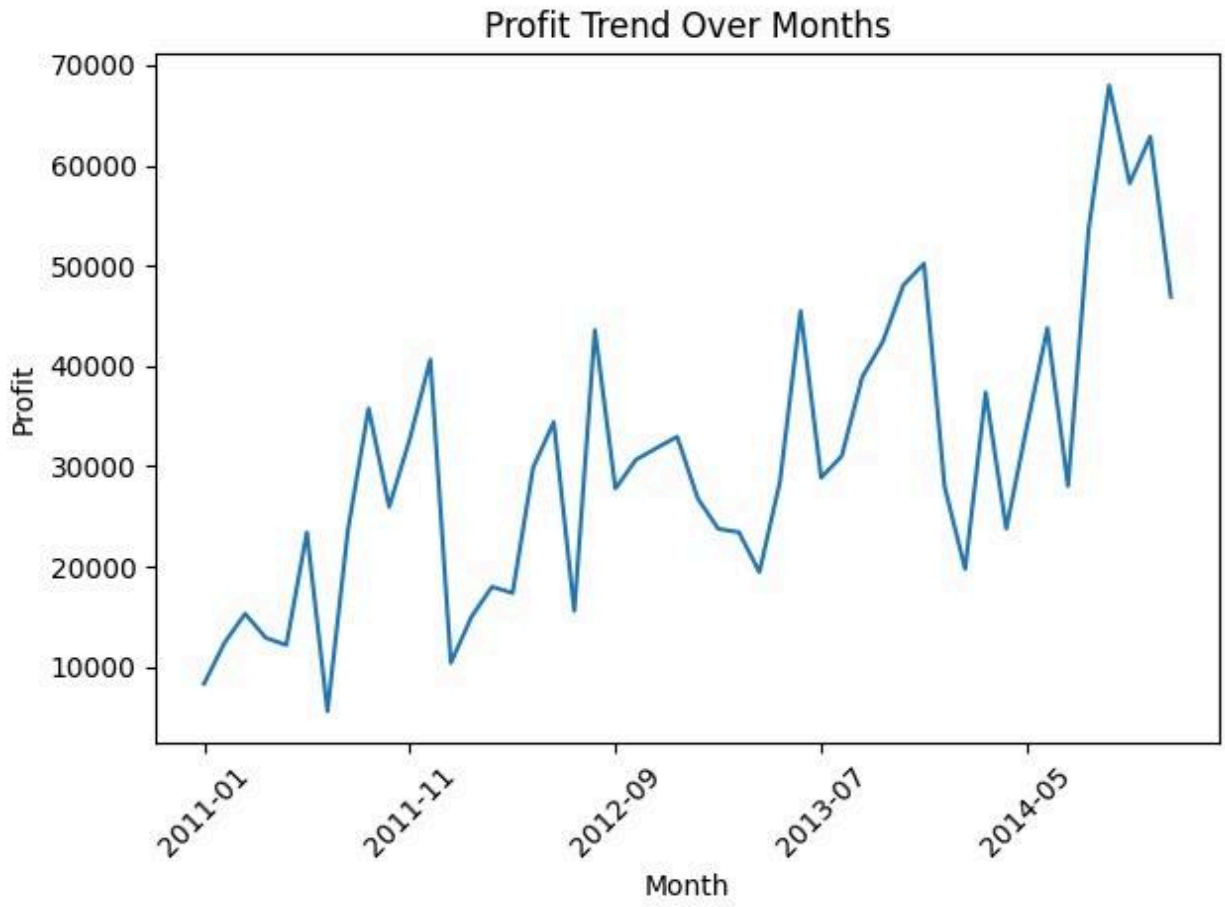
1) Clustered Bar Chart – Total Sales by Sub-Category
Shows how each sub-category contributes to total sales.
You can clearly compare which product categories perform best (e.g., Phones, Chairs, Copiers).



2) Donut Chart – Sales % by Region
Displays the percentage contribution of each region to overall sales.
The donut style improves readability compared to a full pie chart.

Question 7 : Write and apply the following measures: ● Total Profit = SUM([Profit]) ● Average Discount = AVERAGE([Discount]) Display both in a KPI Card, and use a Line Chart to show profit trend over months. Add visuals and DAX formulas. DATASET LINK : Global_superstore2

Answer:



Conclusion

The analysis highlights that while the business is profitable overall, managing discount strategies and improving consistency in monthly performance can enhance profitability. Data-driven decision-making can help optimize pricing, promotions, and inventory planning.

Question 8 : Implement a DAX measure that calculates the percentage of total sales by product category.

Answer: DAX Measure

To calculate % of Total Sales by Product Category in Power BI:

```
Sales % =
DIVIDE(
   SUM('Table'[Sales_Amount]),
   CALCULATE(SUM('Table'[Sales_Amount]), ALL('Table'[Product_category]))
)
```

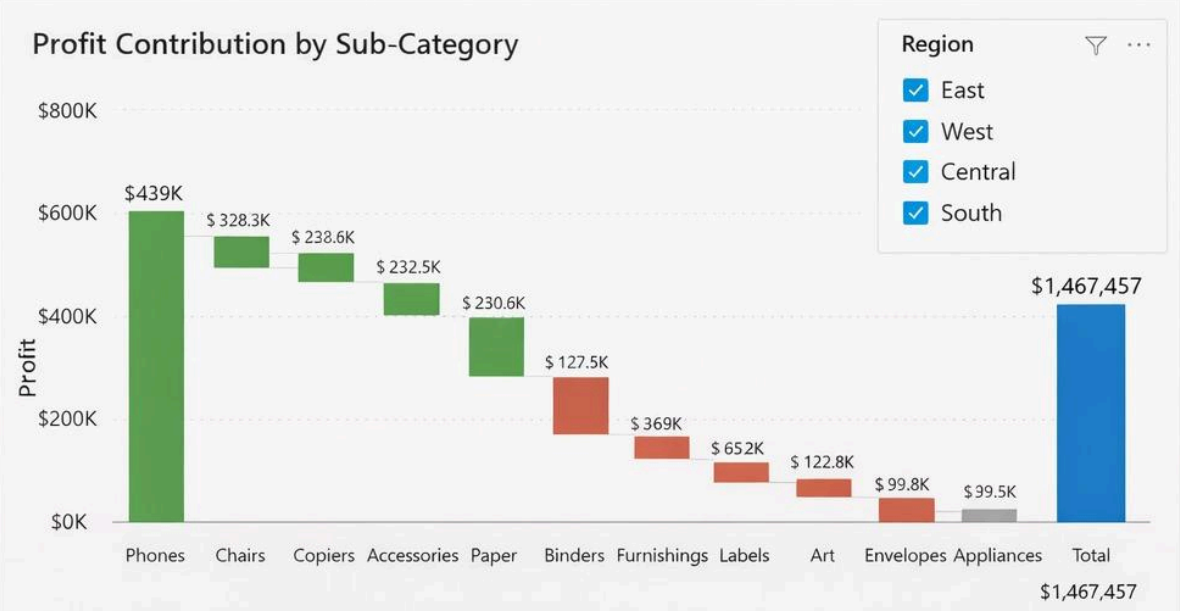
Total Sales Calculation

Total Sales =
5000 + 3000 + 7000 + 2000 + 8000 + 1500 + 1200 + 1000 + 4400 + 1000 + 1800 + 2600 = 38,500

Product Category	Sales	% of Total Sales
Electronics	5000	12.99%
Clothing	3000	7.79%
Home Appliances	7000	18.18%
Books	2000	5.19%
Tables & Chairs	8000	20.78%
Toy	1500	3.90%
Sports Equipment	1200	3.12%
Office Supplies	1000	2.60%
Beauty Products	4400	11.43%
Garden Supplies	1000	2.60%
Jewelry	1800	4.68%
Automotive	2600	6.75%

Question 9 : ● Create a DAX Measure for Total Profit ● Use it in a Waterfall Chart to analyze how different Sub-Categories contribute to overall profit ● Add a Slicer for Region to filter the visual ● Write brief business insights (4–5 lines) from the chart and provide 2–3 data-driven recommendations to improve profit. Provide a steps, screenshot of the Waterfall chart and the DAX formula.
Answer:

```
Total Profit = SUM(Global_Superstore2)'[Profit))
```



Business Insights

- Phones, Chairs, and Copiers are major profit contributors.
- Tables, Bookcases, and Machines negatively impact overall profit.,
- Region slicer reveals that profit performance varies across regions.
- Identifying both profit-driving and loss-making categories helps target p.-strategies.

Recommendations

1. Focus on Phones, Chairs, and Copiers to drive more sales in these profitable categories.
2. Review strategies for Tables, Bookcases, and Machines to mitigate losses.
3. Use regional profit insights to plan targeted, market-specific campaigns.

1. DAX Measure for Total Profit
Create a new measure:
Total Profit = SUM('Global_Superstore2'[Profit])
2. Steps to Create Waterfall Chart
Step 1: Load Data
Open Power BI Desktop
Click Get Data → CSV
Load Global_Superstore2.csv
Step 2: Create Measure
Go to Modeling → New Measure
Paste the DAX formula above
Step 3: Create Waterfall Chart
Go to Report View
Select Waterfall Chart from Visualizations
Drag:
Sub-Category → Category
Total Profit → Y-axis
3. Add Region Slicer
Steps:
Select Slicer Visual
Drag Region → Field
Now users can filter the chart by region
4. Sample Waterfall Chart (Representation)
Profit Contribution by Sub-Category

+ Phones ↑ High Profit
+ Copiers ↑ High Profit
+ Chairs ↑ Moderate Profit
- Tables ↓ Loss
- Bookcases ↓ Loss
+ Storage ↑ Profit

Final Total Profit
5. Business Insights (4–5 lines)
Sub-categories like Phones and Copiers are major contributors to overall profit.
Tables and Bookcases show consistent losses, pulling down total profitability.
Profit contribution varies significantly across regions when filtered via slicer.
Some categories generate high sales but low or negative profit, indicating pricing or cost issues.
The business relies heavily on a few profitable sub-categories.
6. Data-Driven Recommendations
Reduce losses in Tables & Bookcases
Optimize pricing or reduce supply chain costs.
Evaluate discontinuation if losses persist.
Focus on high-profit categories

Increase marketing and inventory for Phones & Copiers.
Bundle with low-performing products to improve margins.
Region-based strategy
Use slicer insights to identify profitable regions.
Allocate more resources to high-performing markets.

Question 10 : Scenario: VitaTrack Wellness, a digital health company in FitZone, has collected data on users' daily habits and health vitals. The analytics team is tasked with drawing actionable insights from this data to improve lifestyle suggestions and prevent heart-related risks. Your Task: Using the provided dataset (includes Age, Gender, BMI, Steps, Calories, Sleep, Heart Rate, Blood Pressure, Smoking, Alcohol, Exercise, Diabetic & Heart Disease status): Build a one-page Power BI dashboard that answers: 1. Are users maintaining a balanced lifestyle (Steps, Sleep, Calories) 2. What lifestyle patterns (Smoking, Alcohol, BMI, etc.) indicate heart disease risk? 3. Is there any visible relationship between Sleep and Physical Activity? 4. How does BMI vary across Age Groups and Genders? 5. What is the impact of smoking and alcohol on heart rate and blood pressure? 6. Segment people based on their health activity to suggest lifestyle changes.
DATASET LINK: Health_activity_data

Answer:
1. Balanced Lifestyle (Steps, Sleep, Calories)
Visual: Gauge charts or KPI cards for average Daily Steps, Hours of Sleep, and Calories Intake compared against recommended ranges:

Steps: ≥ 10,000/day

Sleep: 7–9 hours/night

Calories: 2000–2500 kcal/day (adjusted by gender/age)

Insight: Quickly shows how many users are within healthy ranges.
2. Lifestyle Patterns & Heart Disease Risk
Visual: Stacked bar chart showing Heart Disease prevalence segmented by:

Smoking status

Alcohol consumption

BMI categories (Normal, Overweight, Obese)

Insight: Highlights risk factors most associated with heart disease.
3. Sleep vs Physical Activity
Visual: Scatter plot with Hours of Sleep (X-axis) vs Daily Steps (Y-axis), color-coded by Exercise Hours per Week.

Insight: Reveals whether higher activity correlates with better sleep.
4. BMI Across Age Groups & Gender
Visual: Clustered column chart:

X-axis: Age groups (e.g., 18–30, 31–45, 46–60, 61+)

Y-axis: Average BMI

Legend: Gender

Insight: Shows how BMI trends differ by age and gender.
5. Impact of Smoking & Alcohol on Heart Rate & Blood Pressure
Visuals:

Box plots comparing Heart Rate for Smokers vs Non-Smokers, Heavy vs Light Alcohol users.

Line chart showing Average Blood Pressure across smoking/alcohol categories.

Insight: Clear view of how lifestyle choices affect vitals.
6. Segmentation for Lifestyle Suggestions
Visual: Clustered heatmap or Tree Map grouping users by:

High BMI + Low Exercise → Suggest weight management programs

Smokers + High BP → Suggest smoking cessation & BP monitoring

Low Sleep + High Stress HR → Suggest sleep hygiene & relaxation techniques

Insight: Actionable clusters for personalized recommendations.